

NDC Web Application Documentation

This web application is a data management platform built with Node.js, Express, PostgreSQL, EJS, and various middleware. It features user authentication, file uploads, activity logging, event data handling, and the ability to export data in CSV and Excel formats.

/uploads	-> Uploaded user images
/public	-> Static files (CSS, JS)
/views	-> EJS templates
.env	-> Environment variables
index.js	-> Main application file (your provided code)

Technologies Used

- Node.js & Express: Backend server
- EJS: Templating engine
- PostgreSQL: Relational database
- Multer: File uploads
- ExcelJS: Excel file generation

- json2csv: CSV export
- express-session: Session handling
- dotenv: Environment variable management

Application Setup

1. Install Dependencies

bash

```
npm install
```

2. Start the Server

bash

```
node index.js
```

The app will run at: <http://localhost:3300>

Database Schema Overview

Users Table

sql

CopyEdit

```
CREATE TABLE users (
```

```
id TEXT PRIMARY KEY,  
password TEXT,  
email TEXT UNIQUE,  
image TEXT,  
user_name TEXT,  
role TEXT  
);
```

Data Table (for events)

sql

CopyEdit

```
CREATE TABLE data (  
id TEXT PRIMARY KEY,  
day DATE,  
minute INT,  
second INT,  
latitude FLOAT,  
longitude FLOAT,  
h FLOAT,  
mb FLOAT,  
ml FLOAT,  
az FLOAT,  
location TEXT,  
nearest_location TEXT  
);
```

Activity Logs Table

sql

CopyEdit

```
CREATE TABLE activity_logs (  
  id SERIAL PRIMARY KEY,  
  user_id TEXT,  
  user_email TEXT,  
  action TEXT,  
  details TEXT,  
  timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```



Authentication & Authorization

- `passcode` middleware validates login credentials.
- `isAuthenticated` middleware restricts pages to logged-in users.
- `requireAdmin` middleware restricts certain routes to admins only.
- Sessions are used to persist user login.

Key Routes

Home/Login

- **GET /** → Home page
- **POST /submit** → Authenticate and login

Dashboard & User Pages

- **GET /dashboard** → Dashboard view
- **GET /user** → Admin user panel
- **GET /add** → Form to add user or event

User Management

- **POST /add** → Add new user (admin only)
- **GET /logout** → Log out current user

Event Data

- **POST /events** → Add or update event
- **GET /manage-data** → Manage events
- **DELETE /data/:id** → Delete specific event
- **GET /search?query=** → Search event data

Export Data

- **GET /download** → Download all events as CSV
- **GET /download-excel** → Download all events as Excel

Activity Logs

- **GET /activities** → View all logs (admin only)



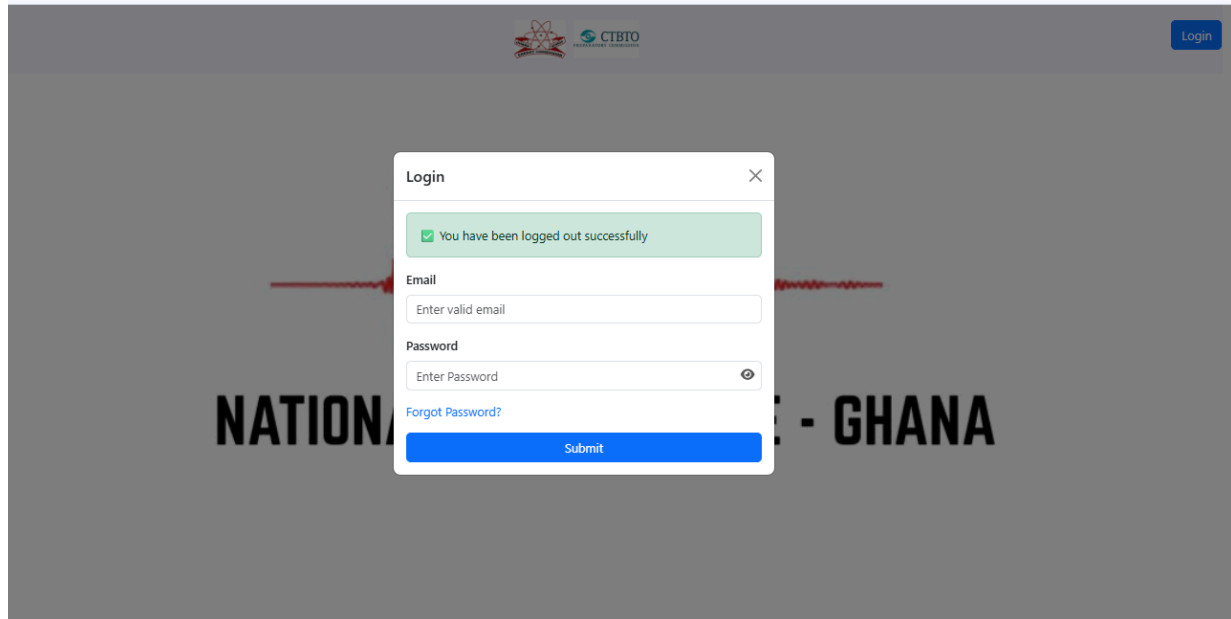
Features

-  User login with session-based authentication

-  Admin role protection
 -  File upload (user image)
 -  Dashboard with metrics (users, staff, etc.)
 -  Event data entry with conflict resolution (upsert)
 -  Event data deletion
 -  Search functionality
 -  Export to **CSV** and **Excel**
 -  Activity logging for login, logout, data operations
-

Validation & Error Handling

- Ensures all required fields are submitted.
- Logs errors to the console.
- Returns relevant status codes for client issues (400, 403, 500, etc.)



Key Elements of the Interface:

1. Header Section:

- **Logos:**

- Left: Logo of the **Ghana Atomic Energy Commission**.
- Right: Logo of the **CTBTO Preparatory Commission**.

- **Center Image:** A rotating globe or Earth image representing a global or scientific context to display all three events being done by the ctbto

Login Form Components

Centered in a white rounded card with shadow effect:

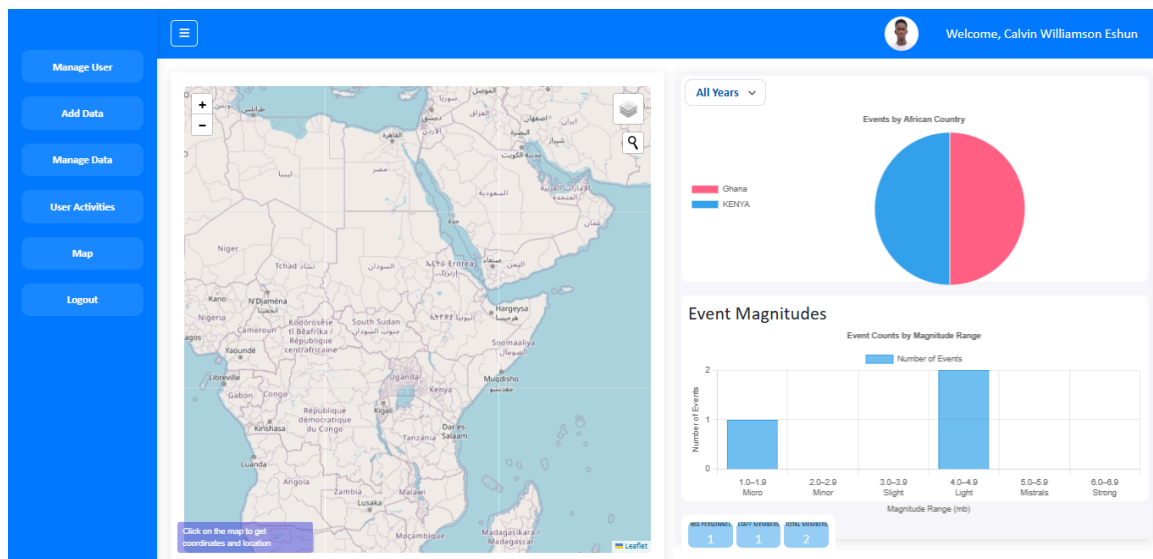
- **Title:**
 - **LOGIN** in bold blue text.
- **Input Fields:**
 - **EMAIL** – With placeholder: “Enter valid email”.
 - **PASSWORD** – With placeholder: “Enter Password” and masked input.
- **Submit Button:**
 - Bright blue with white text: SUBMIT
 - Well-styled and centered for user accessibility.

Visual Background

- **Header Image:** image of Earth overlaid with some satellite imagery.
- **Main Background:** Blurred photo of a modern institutional building with well-maintained green lawns (possibly the headquarters of the Ghana Atomic Energy Commission).

Footer

- Centered copyright:
 - © 2025 Ghana Atomic Energy Commission



This image shows a dashboard interface for a web application, likely used for administrative management at the Ghana Atomic Energy Commission (GAEC) or a related institution.

◆ Dashboard Overview

User Profile Section (Center-Top)

- Profile : Using the image of the user
- USERS NAME: **CALVIN WILLIAMSON ESHUN**
- **Header Greeting (Top-Right Corner):**
 - Profile thumbnail and greeting: *"Welcome, Calvin Williamson Eshun"*



Statistics / Summary Cards (Center-Main)

Each card has a sea-blue background with rounded corners and includes both a label and value:

1. NSS Personal

- Value: **2**

2. Staff Members

- Value: 1

3. Total Members

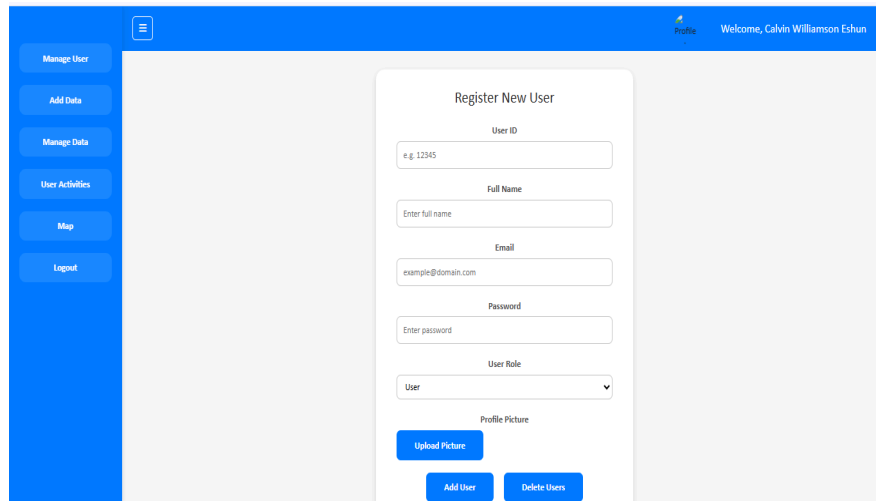
- Value: 3

Sidebar Navigation (Left Panel)

Bright blue vertical sidebar with white text buttons:

1. Manage User
2. Add Data
3. Manage Data
4. User Activities
5. Map
6. Logout

These provide quick access to core dashboard functionalities.



The screenshot displays a web dashboard with a blue sidebar on the left containing navigation links: 'Manage User', 'Add Data', 'Manage Data', 'User Activities', 'Map', and 'Logout'. The main content area features a 'Register New User' form. The form includes input fields for 'User ID' (with a placeholder 'e.g. 12345'), 'Full Name' (with a placeholder 'Enter full name'), 'Email' (with a placeholder 'example@domain.com'), and 'Password' (with a placeholder 'Enter password'). Below these is a 'User Role' dropdown menu currently set to 'User'. At the bottom of the form is an 'Upload Picture' button. Below the form are two buttons: 'Add User' and 'Delete Users'. The top right of the dashboard shows a 'Profile' icon and a welcome message: 'Welcome, Calvin Williamson Eshun'.

User Registration

Overview

This document describes the steps and logic for registering a new user in the NDC-GAEC system. The registration functionality ensures that new users can be added securely, with their credentials stored properly and validated for integrity.

Access Control

- Access Level: Admin only
 - Route: `/user` (or `/register`, depending on implementation)
 - Authentication Required: ☒ Yes (Admin login required)
-



Registration Form Fields

Field Name	Input Type	Required	Validation Rules
Full Name	Text	☒ Yes	Alphabetical only
Email Address	Email	☒ Yes	Must be valid email format, unique
ID Number	Number/Text	☒ Yes	Must be unique
Password	Password	☒ Yes	Min 8 characters, 1 capital, 1 digit
Role	Dropdown	☒ Yes	E.g., Admin, Staff, NSS

Profile Image	File	Optional	JPG/PNG, max size 2MB
---------------	------	----------	-----------------------

Server-Side Logic

Route: **POST** /user

1. Receive Input

The backend receives form data via `req.body` and uploaded image via `req.file`.

2. Validation

- Check required fields
- Validate email format
- Confirm password match
- Check uniqueness of email and ID

3. Save to Database

- Insert the new user into the `users` table with the password

- Save image filename if uploaded

4. Log Activity (optional)

- Record the registration action in a `user_activity_log` table

5. Send Response

- Redirect to dashboard or display success message

Success Responses

- Message: "Registration successful. User has been added."
- Redirect: `/dashboard` and the new user can access the website successfully

DELETE USER SECTION

This is managed by only the Admin, whereas the admin is allowed to delete a user from the system, this allows efficiency and clear cached data

✖ Error Responses

Error Type	Message
Validation Error	"Email is already registered."
Input Missing	"All required fields must be filled."

Manage User

Add Data

Manage Data

User Activities

Map

Logout

Upload Data ▾

SHI Data

ID	Year	Day	Month	Minute	Second	Hour	Latitude	Longitude	H	Mb	MI	Az	Location	Nearest Location
Event ID	0000	DD	MM	00:00	00:00	HR								

Submit Event

Radionuclide Data

Day	Year	Month	Minute	Seconds	Hour	Latitude	Longitude	Depth	Mb	MI	Az	Location	Nearest Location
DD	0000	MM	00:00	00:00	HR								

Submit Event

Add Data & Display Records

Overview

This module allows authorized users to add new records (e.g., personnel, staff, NSS members) to the database and view all saved entries. The system ensures that data integrity is maintained and entries can be reviewed efficiently in a tabular format.

Access Control

- **Access Level:** Authenticated users (e.g., admin, staff and national service personnels)
- **Routes:**
 - GET /add → Display add-data form
 - POST /add → Handle form submission and save to DB
 - GET /manage-data → Display all data entries in a table

<

Manage Data

Overview

The Manage Data module allows authorized users to view, search, and delete records from the system. This functionality is essential for maintaining data accuracy and ensuring the removal of incorrect or duplicate entries.

Access Control

- Access Level: Authorized users and Admins
- Route: </manage-data>

- Authentication Required:  Yes
-

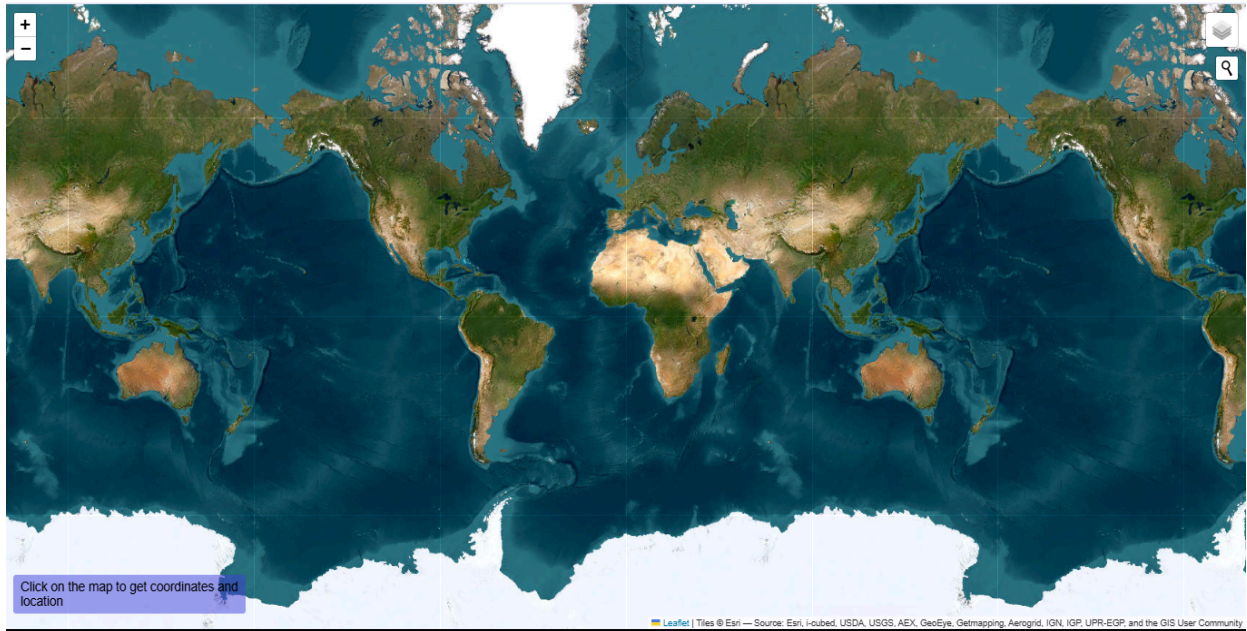
Functional Description

Features

- Display all records in a table format
- Uploading data in excel format
- Downloading saved template to help organise your upload and prevent errors
- Search/filter records (optional enhancement)
- Delete a record if the input was incorrect

Download field

Has a download section at the top where users can download data in csv format or excel format to work manually with.



Overview

The Map Page allows users to search for a specific location and view its latitude and longitude. This tool is useful for visualizing user-submitted coordinates, site visits, or planning geolocation-based data entries.

Key Features

Feature	Description
Search Bar	Input field that accepts a location name (city, region, or coordinates)

Map Display	Interactive map rendered via a map library (e.g., Leaflet)
Coordinate Display	Dynamic text fields that show the selected location's latitude and longitude
Marker Pin Drop	Clicking on the map places a marker and updates coordinates accordingly

How It Works

1. Search Functionality

- Users type a location (e.g., "Accra, Ghana") in the search bar.
- The map centers on the searched location using a geocoding API (e.g., OpenStreetMap Nominatim or Google Maps Geocoding).
- A marker is dropped on the result.

2. Click to Get Coordinates

- When the user clicks anywhere on the map, the system drops a marker at the clicked point.
- Latitude and longitude are displayed in real time in the designated input field.

USER ACTIVITY LOGS				
User id	Email	Action	Details	Time
5	kvekujur143@gmail.com	View Dashboard	User accessed dashboard	Thu Jul 10 2025 13:04:24 GMT+0000 (Greenwich Mean Time)
5	kvekujur143@gmail.com	Login	User logged in successfully	Thu Jul 10 2025 13:04:24 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	Logout	User logged out	Thu Jul 10 2025 13:01:48 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	View Dashboard	User accessed dashboard	Thu Jul 10 2025 13:01:44 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	Login	User logged in successfully	Thu Jul 10 2025 13:01:41 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	View Dashboard	User accessed dashboard	Thu Jul 10 2025 13:01:41 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	Login	User logged in successfully	Thu Jul 10 2025 13:01:41 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	Logout	User logged out	Thu Jul 10 2025 13:01:37 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	View Dashboard	User accessed dashboard	Thu Jul 10 2025 13:01:31 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	Login	User logged in successfully	Thu Jul 10 2025 13:01:31 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	Logout	User logged out	Thu Jul 10 2025 13:01:13 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	View Dashboard	User accessed dashboard	Thu Jul 10 2025 12:56:25 GMT+0000 (Greenwich Mean Time)
2	jeffery@gmail.com	Login	User logged in successfully	Thu Jul 10 2025 12:56:25 GMT+0000 (Greenwich Mean Time)

User Activities Section

Overview

The User Activities section provides administrators with a log of all critical actions performed by users within the system. This feature supports auditing, accountability, and security monitoring, allowing admins to trace actions back to individual users.

Access Control

- Access Level: Admin only
 - Route: /activities
 - Authentication Required: ☒ Yes (Admin login required)
-

Purpose

The section aims to:

- Monitor system usage and user behavior
- Provide transparency for sensitive operations
- Detect unauthorized or unusual activities
- Keep historical records of user interactions